

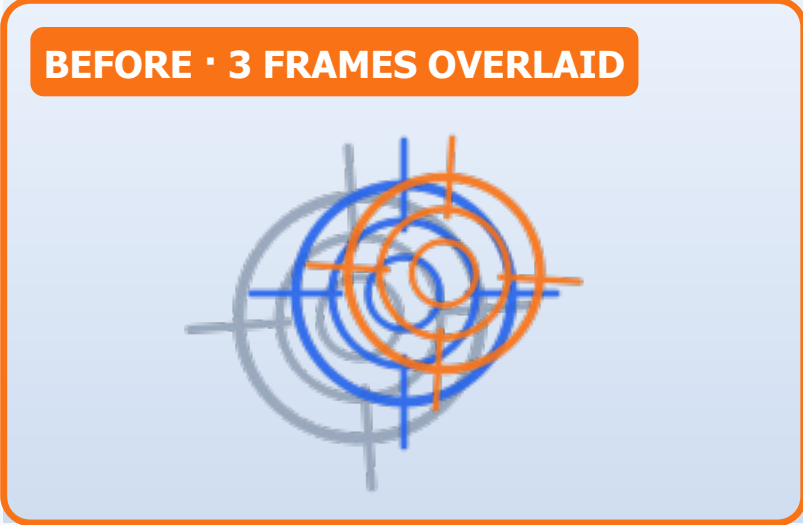
Image Registration Procedures

Focus-breathing correction · feature homography · ECC refinement · common-region crop



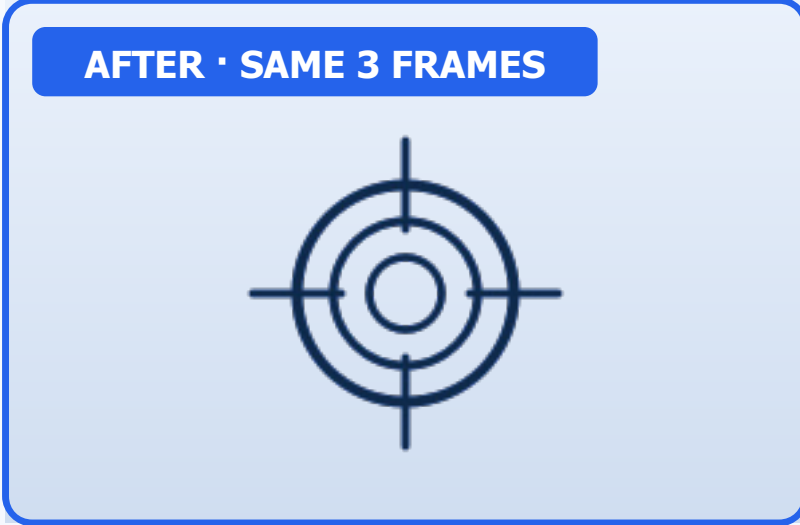
1 Why the stack has to be aligned first

Fusion decides pixel by pixel. If the frames do not sit on the same geometry, it fuses detail from places that never lined up.



Magnification and centre drift frame to frame.

register



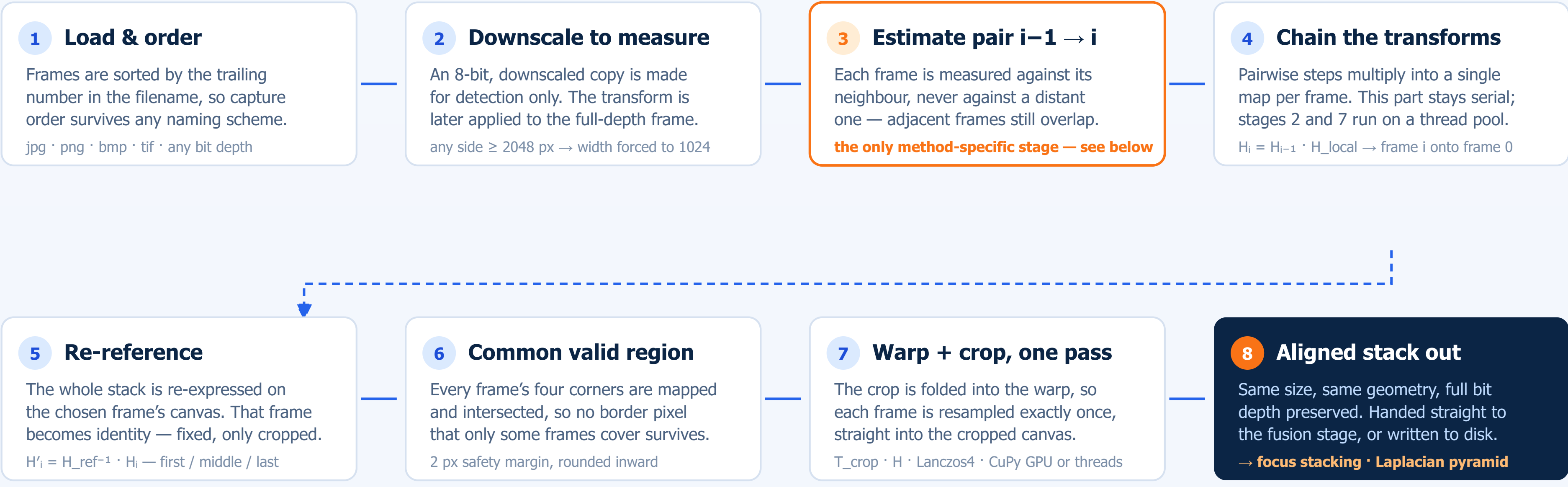
One geometry — every frame on the same grid.

What moves between frames

- **Focus breathing.** Refocusing changes magnification — a uniform scale about the optical axis, largest at close focus.
- **Rail and mount drift.** Translation, small rotation, and on a tilted subject a genuine perspective change.
- Left uncorrected, both reconstruct as doubled edges and haloes.

2 The procedure — eight stages, identical for every method

Only stage 3 differs between scale, homography and ECC. Everything around it — the chain, the re-reference, the crop — is shared.



! If a frame yields too few features (< 4 keypoints) or too few good matches (< 6), its own estimate is discarded and the running trajectory is reused — the stack never breaks, it just stops correcting for that one frame.

3 Stage 3 in four flavours — how the pair transform is measured

More degrees of freedom is not better. A model looser than the real motion will fit the blur difference instead of the misalignment.



4 Two choices that decide the output

Which frame stays fixed, and how much of the canvas every frame can agree on.

